KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

MATHEMATICS

Year 9 5.2

Yearly Exam

2026

Instructions

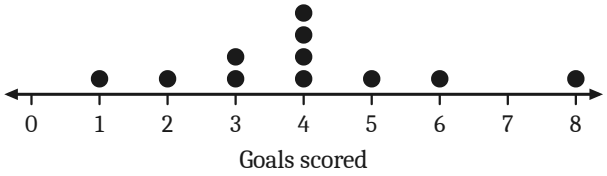
1. Reading time: 5 minutes
2. Working time: 90 minutes
3. Attempt all questions
4. Show ALL working where needed
5. A calculator is permitted

Sections and Marks

Section 1 - Data	/7
Section 2 - Surface area and volume	/7
Section 3 - Mixed: Chapters 1-8	/6
Total marks	**/20**

Section 1: Data

Multiple Choice:

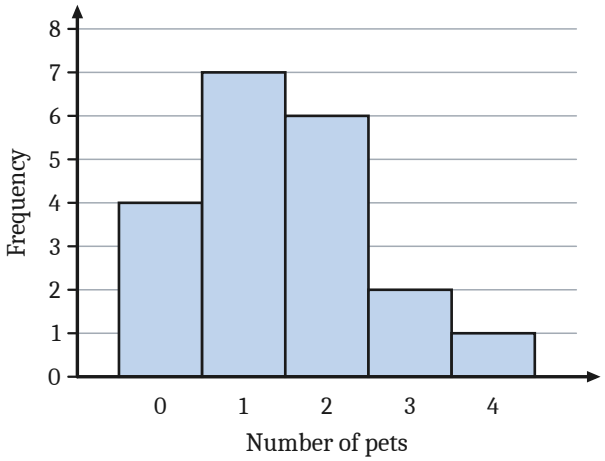
		Mark
1	[PLACEHOLDER] The dot plot shows the number of goals scored by a netball team in each game. What is the mode? A. 7 B. 4 C. 5 D. 11 	1

Short Answer:

	Question	Working Out	Mark
2	[PLACEHOLDER] The stem-and-leaf plot shows the ages of the people at a family lunch. Stem | Leaf 2 | 3 5 3 | 1 4 4 8 4 | 2 5 7 5 | 1 6 Key: 2 | 3 = 23 a. Find the median age. b. Find the range.	a. b.	1 1

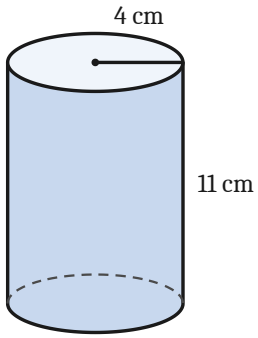
Section 1: Data (continued)

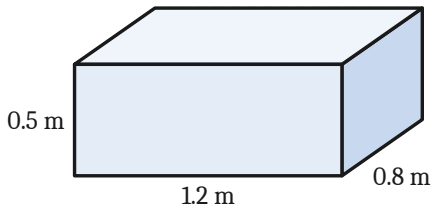
Worded Problems:

	Question	Working Out	Mark
3	[PLACEHOLDER] A group of Year 9 students were asked how many pets they have. The results are shown in the frequency table and the histogram. **Pets (x) / Frequency (f)** 0 — 4 1 — 7 2 — 6 3 — 2 4 — 1 		
a	How many students were surveyed?		1
b	What is the mode?		1
c	Find the mean number of pets, correct to 1 decimal place.		2
	Total		**/7**

Section 2: Surface area and volume

	Multiple Choice:	Mark
1	[PLACEHOLDER] What is the volume of a cube with sides of 5 cm? A. 15 cm^3 B. 25 cm^3 C. 125 cm^3 D. 150 cm^3	1

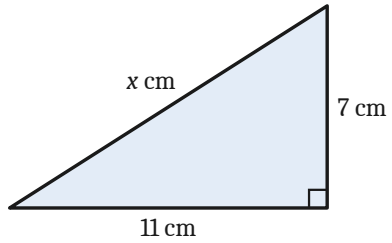
	Short Answer:		
	Question	**Working Out**	**Mark**
2	[PLACEHOLDER] Find the volume of this cylinder, correct to 1 decimal place.  Note: Volume of a cylinder: $V = \pi r^2 h$		2

	Worded Problems:		
	Question	**Working Out**	**Mark**
3	[PLACEHOLDER] A fish tank is a rectangular prism with an open top.  		
a	Find the volume of the tank in m^3.		1
b	How many litres of water does the tank hold? ($1\ m^3 = 1000$ L)		1

	Section 2: Surface area and volume (continued)		
	Question	**Working Out**	**Mark**
c	Glass for the tank costs \$45 per m^2. Find the area of glass needed (no lid) and its cost.		2
	Total		**/7**

Section 3: Mixed: Chapters 1-8

	Multiple Choice:	Mark
1	[PLACEHOLDER] Simplify $a^5 \times a^3$. A. a^2 B. a^8 C. a^{15} D. $2a^8$	1

	Short Answer:		
	Question	**Working Out**	**Mark**
2	[PLACEHOLDER] Find the value of x, correct to 1 decimal place.  Note: Pythagoras' theorem: $c^2 = a^2 + b^2$		2

	Worded Problems:		
	Question	**Working Out**	**Mark**
3	[PLACEHOLDER] Mia works at a garden centre for $24.60 per hour. Last week she worked 38 hours at the normal rate and 4 hours of overtime at time-and-a-half.		
a	Find her pay for the normal hours.		1
b	Find her total pay for the week.		2
	Total		**/6**